

Business Intelligence & Analytics

Mock Exam 1 | Course ID: noc26_cs64

Q1. According to the lecture, what is the primary reason modern technology giants like Google and Facebook are valued in the billions?

- A) For the physical infrastructure they build.
- B) For the data they collect and the intelligence they derive from it.
- C) For their traditional advertising revenue models.
- D) For their ability to offer free services without monetization.

Source: Lecture_W1L25_INTRODUCTION_TO__BUSINESS_INTELLIGENCE__AND_ANALYT | Difficulty: Easy

Q2. In the context of Business Intelligence (BI), what is the main purpose of converting raw data into useful information and actionable intelligence?

- A) To reduce the overall volume of data stored.
- B) To support informed decision-making.
- C) To automate all business processes.
- D) To replace human intuition entirely.

Source: Lecture_W1L25_INTRODUCTION_TO__BUSINESS_INTELLIGENCE__AND_ANALYT | Difficulty: Easy

Q3. What is the primary function of Descriptive Analytics?

- A) To predict future outcomes.
- B) To answer the question, "What is happening?"
- C) To establish causation between variables.
- D) To implement solutions based on discovered patterns.

Source: Lecture_W1L26_PATTERNS_IN_DATA | Difficulty: Easy

Q4. The "Long Tail" phenomenon is primarily driven by which two forces?

- A) Limited inventory and high production costs.
- B) Infinite inventory (supply-side) and search/discovery tools (demand-side).
- C) Traditional brick-and-mortar retail and local advertising.
- D) Pareto Principle and mass market appeal.

Source: Lecture_W1L26_PATTERNS_IN_DATA | Difficulty: Hard

Q5. What does the course identify as the "heart" of Business-to-Analytics Translation?

- A) Designing new machine learning algorithms.
- B) Converting a purely business-related problem into a well-defined analytics problem.
- C) Mastering open-source software tools like Python and R.
- D) Building a robust BI&A architecture.

Source: Lecture_W1L27_COURSE_OVERVIEW | Difficulty: Medium

Q6. Which of the following is NOT considered one of the three essential pillars for success in analytics, as described in the lecture?

- A) Technology
- B) Science
- C) Domain Knowledge
- D) Financial Capital

Source: Lecture_W1L28_VOCABULARY_OF_BUSINESS_ANALYTICS | Difficulty: Easy

Q7. What is the primary purpose of ETL (Extraction, Transformation, and Loading) in the analytics pipeline?

- A) To perform real-time transaction processing.
- B) To store unstructured data in its native format.
- C) To ensure data is clean, consistent, and ready for analysis.
- D) To visualize data for end-users.

Source: Lecture_W1L28_VOCABULARY_OF_BUSINESS_ANALYTICS | Difficulty: Easy

Q8. In the AT&T Bizocity case, what was the primary data source that Bell Labs leveraged to identify business prospects, replacing AT&T's previous method of buying competitor directories?

- A) Social media data.
- B) Call Detail Data (CDD) from AT&T's own network.
- C) Publicly available census data.
- D) Customer survey responses.

Source: Lecture_W2L29_CASE_Bizocity_scoring_at_AT_T | Difficulty: Easy

Q9. What is the key distinction between OLTP (Online Transaction Processing) systems and a Data Warehouse in a BI architecture?

- A) OLTP systems are for analysis, while Data Warehouses are for transactions.
- B) OLTP systems are optimized for efficient data capture and storage, while Data Warehouses are optimized for analysis and querying.
- C) OLTP systems handle historical data, while Data Warehouses handle real-time data.
- D) OLTP systems are used by data scientists, while Data Warehouses are used by business users.

Source: Lecture_W2L30_BUSINESS_INTELLIGENCE_ARCHITECTURE | Difficulty: Medium

Q10. Which concept ensures that every record in a database table can be uniquely identified and prevents duplicate entries?

- A) Referential Integrity
- B) Data Redundancy
- C) Entity Integrity
- D) Updation Anomaly

Source: Lecture_W2L31_DATA_MANAGEMENT | Difficulty: Medium

Q11. In relational database design, how is a many-to-many (N-to-N) relationship between two entities typically resolved?

- A) By adding a foreign key to both tables.
- B) By creating an associative entity (junction table).
- C) By merging the two tables into one.
- D) By using a primary key from one table as a foreign key in the other.

Source: Lecture_W2L32_ONLINE_TRANSACTION_PROCESSING | Difficulty: Hard

Q12. Which SQL command is used to combine rows from two or more tables based on a related column between them?

- A) `SELECT`
- B) `WHERE`
- C) `GROUP BY`
- D) `JOIN`

Source: Lecture_W3L69_INTRODUCTION_TO_SQL | Difficulty: Easy

Q13. What is an "Updation Anomaly" in database design?

- A) The inability to add new data without adding unrelated data.
- B) The unintended loss of data when a record is deleted.
- C) Data inconsistencies arising from updating the same information in multiple locations.
- D) The repetition of the same data in multiple places.

Source: Lecture_W3L70_NORMALISATION | Difficulty: Medium

Q14. In the ShopSense case, when calculating quarter-wise tax payable, which SQL aggregate function would be most appropriate to sum the tax values?

- A) `AVG()`
- B) `COUNT()`
- C) `SUM()`
- D) `MAX()`

Source: Lecture_W3L71_SHOPSENSE_CASE_IN_MySQL_Workbench | Difficulty: Easy

Q15. Which OLAP operation involves navigating from a higher-level summary to more detailed data within a dimension (e.g., from Year to Quarter to Month)?

- A) Slice
- B) Dice
- C) Drill-Down
- D) Roll-Up

Source: Lecture_W3L72_ONLINE_ANALYTICAL_PROCESSING | Difficulty: Easy

Q16. Why is customer retention often considered a high-impact strategic priority, even if new customers initially represent a loss?

- A) New customers are always more profitable than old ones.
- B) Tenured customers typically increase purchases, require less support, and provide valuable referrals.
- C) Acquisition costs are usually negligible.
- D) Long-term customers are easier to replace.

Source: Lecture_W4L73_DESCRIPTIVE_DATA_ANALYTICS | Difficulty: Medium

Q17. In churn analysis, what does the "Hazard" rate represent?

- A) The cumulative probability that a customer will remain active up to a given tenure.
- B) The total number of customers who churned in a given period.
- C) The probability that a customer, having survived up to time t , will churn *before* time $t+1$.
- D) The average tenure of all customers in a cohort.

Source: Lecture_W4L74_CHURN_ANALYSIS | Difficulty: Hard

Q18. What is the primary purpose of the "Discount Rate (i)" in the Customer Lifetime Value (CLV) formula?

- A) To account for the cost of customer acquisition.
- B) To adjust for the probability of customer churn.
- C) To reflect the time value of money, making future profits less valuable than present ones.
- D) To measure the customer's past spending habits.

Source: Lecture_W4L75_CUSTOMER_LIFETIME_VALUE | Difficulty: Medium

Q19. According to the CLV spreadsheet analysis, which lever had the largest percentage impact on increasing the 5-year CLV?

- A) Increasing the annual survival rate.
- B) Reducing the acquisition cost.
- C) Increasing the average order value.
- D) Decreasing the discount rate.

Source: Lecture_W4L76_NPV___CLV_SPREADSHEET_ANALYSIS | Difficulty: Hard

Q20. What is the primary reason the "Thought Process" is considered the most critical phase in the Analytics Lifecycle?

- A) It involves collecting all necessary data.
- B) It ensures the entire effort is aimed at solving the *right problem* and delivering a useful output.
- C) It focuses on testing and validating the model.

D) It measures the model's impact on business performance indicators.

Source: Lecture_W5L77_ANALYTICS_PROCESS | Difficulty: Medium

Q21. What does a high Variance Inflation Factor (VIF) value (e.g., > 2) primarily indicate in a regression model?

- A) The variable is highly significant.
- B) The model is suffering from overfitting.
- C) The variable is redundant due to multicollinearity with other independent variables.
- D) The data quality for that variable is poor.

Source: Lecture_W5L78_INTRODUCTION_TO_STATISTICAL_LEARNING_AND_DATA_PRE_ | Difficulty: Hard

Q22. If your primary goal in building a model is **explanation**** (to understand the relative importance of drivers), what crucial preprocessing step should you always perform on your variables before modeling?**

- A) Remove all outliers.
- B) Convert all variables to categorical.
- C) Standardize all variables (e.g., using Z-score normalization).
- D) Apply a log transformation to all variables.

Source: Lecture_W5L79_DATA_MINING_PROCESS | Difficulty: Hard

Q23. What is the primary purpose of K-Fold Cross-Validation in predictive modeling?

- A) To reduce the number of features in the dataset.
- B) To estimate the true test error reliably and mitigate the impact of a single train-test split.
- C) To increase the training error of the model.
- D) To simplify the model by reducing its flexibility.

Source: Lecture_W5L80_OVERVIEW_OF_DATA_MINING_TECHNIQUES | Difficulty: Medium

Q24. In the XYZ Bank case, the analyst initially found that online customers appeared more profitable. However, what critical logical fallacy was identified when considering recommending a rebate strategy?

- A) The sample size was too small to draw conclusions.
- B) The analysis showed correlation, but not necessarily causation, between online status and profitability.
- C) The data was outdated and did not reflect current customer behavior.
- D) The bank's profitability skew was not considered.

Source: Lecture_W5L81_ANALYTICS_PROCESS_CASE | Difficulty: Hard

Q25. What is the defining characteristic that distinguishes classification from clustering?

- A) Classification uses numerical data, while clustering uses categorical data.
- B) Classification is an unsupervised learning technique, while clustering is supervised.
- C) Classification assigns objects to predefined classes, while clustering discovers inherent groupings without predefined labels.
- D) Classification is used for prediction, while clustering is used for explanation.

Source: Lecture_W6L93_INTRODUCTION_TO_CLASSIFICATION | Difficulty: Medium

Q26. In a medical diagnosis scenario where missing a disease (False Negative) is catastrophic, which performance metric should be prioritized?

- A) Accuracy
- B) Specificity
- C) Sensitivity (Recall)
- D) Precision

Source: Lecture_W6L94_SCORING_MODELS | Difficulty: Hard

Q27. What does the AUC (Area Under the Curve) of an ROC curve represent?

- A) The optimal classification threshold for the model.
- B) The model's performance at a single, specific threshold.
- C) A single-number summary of the classifier's performance across all possible thresholds.
- D) The trade-off between Precision and Recall.

Source: Lecture_W6L95_CLASSIFIER_PERFORMANCE | Difficulty: Medium

Q28. What is the primary goal of a decision tree algorithm when making a split?

- A) To maximize the number of nodes.
- B) To achieve perfect heterogeneity in child nodes.
- C) To create child nodes that are as homogeneous (pure) as possible with respect to the target variable.
- D) To ensure all attributes are used in the first split.

Source: Lecture_W7L96_DECISION_TREES | Difficulty: Easy

Q29. For both the Gini Index and Entropy measures of impurity in decision trees, what value indicates perfect purity (all items in one class)?

- A) 1.0
- B) 0.5
- C) 0.0
- D) -1.0

Source: Lecture_W7L97_ATTRIBUTE_SELECTION | Difficulty: Easy

Q30. Which ensemble method enhances Bagging by introducing an additional layer of randomness, where at each split, the algorithm only considers a random subset of the available features?

- A) Boosting
- B) Random Forest
- C) K-Means
- D) Hierarchical Clustering

Source: Lecture_W7L98_GROWING_A_DECISION_TREE | Difficulty: Medium

Answer Key

1: B	2: B	3: B	4: B	5: B
6: D	7: C	8: B	9: B	10: C
11: B	12: D	13: C	14: C	15: C
16: B	17: C	18: C	19: C	20: B
21: C	22: C	23: B	24: B	25: C
26: C	27: C	28: C	29: C	30: B